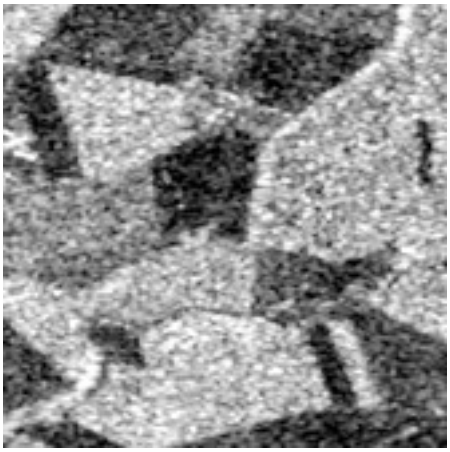
SAR Image Analysis Report



Original SAR

Colorized

K-Means Segmented

Metadata:

- Original File: original_1764397495.png
- Colorized File: colorized_1764397495.png
- Processed On: 2025-11-29 11:55:41 IST
- SAR Classifier Confidence: 0.998
- Colorization Model: Pix2Pix GAN
- KMeans Clusters: 4
- Colorized Resolution: 256 x 256 px

Cluster Analysis:

Cluster 0: Likely Urban / Built-up (Bright/Gray) | Mean RGB: (144, 129, 137) | Area: 41.6%

Cluster 1: Likely Vegetation (Green Tone) | Mean RGB: (11, 27, 25) | Area: 21.48%

Cluster 2: Likely Water (Blue Tone) | Mean RGB: (233, 233, 234) | Area: 8.45%

Cluster 3: Likely Vegetation (Green Tone) | Mean RGB: (70, 94, 84) | Area: 28.46%

Land Cover Summary:

- Likely Urban / Built-up (Bright/Gray) → 41.6%
- Likely Vegetation (Green Tone) → 21.48%
- Likely Water (Blue Tone) → 8.45%
- Likely Vegetation (Green Tone) → 28.46%

Note: Colorized image is GAN-generated and for visualization only.

Analysis Details & Recommendations

Interpretation Notes:

The area is dominated by likely vegetation (green tone) (49.94% combined).

Secondary presence: likely urban / built-up (bright/gray) (41.6%).

Secondary presence: likely water (blue tone) (8.45%).

K-Means groups pixels by color similarity — labels are heuristic.

Consider adding multispectral or time-series data for improved analysis.

Recommendations:

The analysis shows dominant land cover is urban/built-up (41.73%). High urban/built-up fraction indicates potential zones for infrastructure assessment and urban expansion monitoring. The second-largest class is vegetation (28.49%), so combined management strategies addressing both urban/built-up and vegetation are advisable. Recommended actions: collect ground-truth samples where possible, prioritize monitoring based on dominant class, and re-run analysis after incorporating higher-resolution data or additional bands.

Confidence Indicators:

SAR Detection Confidence: 0.998
Dominant Class Percentage: 41.60%
Classification Reliability: High

Limitations:

- KMeans is unsupervised and may not match exact land-cover classes.
- GAN colorization is synthetic and does not represent true spectral values.
- Classification depends on image quality and may contain uncertainty.
- Single-image analysis; use multi-temporal data for trend-based conclusions.

Final Summary:

The analysis indicates predominant vegetation with mixed surfaces.

No major water bodies or dense urban zones detected in this image.

Recommend periodic monitoring and ground-truth sampling to improve confidence.